

JANNATUL MAHMUD

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SUMMARY

Computer Engineering student with hands-on experience in low-level programming (LC-3 assembly), systems-level coding in C++ and Java, embedded development on ESP32, and digital logic design. Strong self-learner who has independently picked up assembly, quantum computing (Qiskit), and embedded systems outside of coursework. Comfortable communicating clearly with both technical and non-technical audiences via email and voice. Seeking a software development role where I can grow into mainframe and systems-level work.

EDUCATION

Wright State University — Fairborn, OH Jan 2026 – Dec 2027
B.S. in Computer Engineering (in progress, ~3 semesters remaining)

Sinclair Community College — Dayton, OH Aug 2023 – Dec 2025
Associate Degree in Computer Science

TECHNICAL SKILLS

- **Languages:** C++, Java, Python, LC-3 Assembly
- **Web:** HTML, CSS, JavaScript (basic)
- **Tools & Environments:** Linux/Unix command line, macOS Terminal, Git/GitHub, VS Code, Eclipse, Visual Studio 2022
- **Hardware & Logic:** Logisim Evolution (digital logic), ESP32-S3 / Arduino, HC-SR04 ultrasonic sensors, multimeter and basic bench wiring
- **Concepts:** Object-oriented programming, data structures (BST, dictionaries), runtime stack and function-call conventions, finite state machines, ROM-based control units, debugging with serial monitors and emulators
- **Soft Skills:** Technical communication, documentation, leadership in student organizations, mentoring peers

PROJECTS

LC-3 Assembly Programs — Computer Organization

LC-3 Assembly | Self-directed coursework

- Wrote multiple low-level assembly programs including a calculator (add/subtract/multiply), a two-digit input reader, and a routine that finds the largest value in an array passed by pointer and count through the runtime stack.
- Implemented JSR/RET function calls with proper register save and restore, frame pointer (R5) and stack pointer (R6) conventions, and TRAP routines for I/O (GETC, OUT, PUTS).
- Debugged real assembler issues including immediate field range errors, branch offset problems, and ASCII-to-digit conversion.

ESP32-S3 Ultrasonic Traffic Light Controller

C++ / Arduino | ESP32-S3-N16R8, HC-SR04 sensor

- Built a working traffic-light controller in C++ on an ESP32-S3 that uses an HC-SR04 ultrasonic sensor to detect cars and switch the lights early when the road clears.
- Designed a 1kΩ / 2kΩ voltage divider on the ECHO line to safely drop the sensor's 5V signal to the ESP32's 3.3V GPIO, preventing pin damage.
- Diagnosed and fixed common embedded issues: serial connection failures, pulseIn timeouts, and baud-rate mismatches.

Logisim Lab 11 / Lab 12 — Data Unit and Control Unit

Logisim Evolution | Digital logic and computer organization

- Designed an 8x4 register file with a 3-to-8 decoder for one-hot write enable, multiplexers for register read selection, and a 4-bit ALU supporting 16 operations including add, subtract (two's complement), multiply, divide, NOT, and shifts.

- Implemented PZN status flags (P = NOT(Z) AND NOT(sign), Z = (F == 0), N = sign bit) and a ROM-based control unit driving microinstructions for branch and sequencing.
- Built a highway / farm-road traffic-light FSM with a 2-bit yellow timer and reset logic using a 2:1 mux in front of the state register.

Java Enhanced Binary Search Tree & C++ Array Dictionary

Java, C++ | Data Structures coursework

- Implemented a generic binary search tree in Java with insert, search, traversal, and balancing-aware operations as part of a Data Structures assignment.
- Built an Array-based Dictionary class in C++ following provided textbook headers, including resize logic, key/value pair management, and full unit-style testing.

Code in Place — Stanford University

Python | April 2023 – June 2023

- Completed Stanford's structured Python program covering fundamentals, problem-solving, and small real-world applications. Earned Code in Place certificate.

EXPERIENCE

Executive Member — WSU Bangladesh Student Association

May 2022 – Present

- Help organize cultural and networking events for international students; mentor new students adjusting to U.S. academic life.
- Coordinate logistics, communication, and outreach across a multi-member executive team.

CERTIFICATIONS & ADDITIONAL

- Code in Place Certificate — Stanford University (2023)
- Blood donor and volunteer with BADHAN (2018 – 2020)
- International student; authorized to work in the U.S. through F-1 / CPT (confirm with employer as needed)